Nonnegativity of the small g-matrix in rank three

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Editorial scope. The rank-three bound is obtained through a source-specific lift to a cylinder, k-arcs of projected great-circle arrangements and a transition-count argument relating the two ends of a motion to the g-matrix. Crossing-number and sublevel applications are retained without separate counting. Complete author mathematical body and appendices retained, title/abstract replaced by delivery wrapper. The native arcs paper has two equation references whose definitions appear in the authors companion2504.07752v1; their original author statements are now appended as explicit CC BY4.0 context, without changing the native paper or mathematical references.

Allowed standard prerequisites (editorial).

- Standard convex-polytope and arrangement theory. Original references retained; all source-specific mutation, transform, and geometric arguments are included.

1 Introduction

In this paper, we study the interplay between several classical topics in discrete and computational geometry: the combinatorial theory of convex polytopes, the complexity of $(\leqslant k)$-sublevels in arrangements, Radon partitions, and crossing numbers. Let $V = \{v_1, \ldots, v_n\} \subset \mathbb{R}^3$ be a set of $n \geqslant 3$ vectors; fixing the labeling of the vectors, we will also view V as an $(3 \times n)$-matrix $V = [v_1 | \ldots | v_n] \in \mathbb{R}^{3\times n}$ with column vectors v_i. Unless stated otherwise, we assume

that V is in general position, i.e., that any 3 of the vectors are linearly independent. We refer to V as a vector configration of rank 3. Let S^2 be the unit sphere in $\mathbb{R}^3$, let $\langle \cdot, \cdot \rangle$ denote the standard inner product in $\mathbb{R}^3$, and let $\mathrm{sgn}(x) \in \{-1, 0, +1\}$ denote the sign of a real number $x \in \mathbb{R}$. For $F \in \{-1, 0, +1\}^n$, let F_+, F_0, and F_- denote the subsets of indices $i \in [n]$ such that $F_i = +1$, $F_i = 0$, and $F_i = -1$.

1.1 Dependency and Dissection Patterns

We consider two sets $\mathcal{F}(V), \mathcal{F}^*(V) \subset \{-1, 0, +1\}^n$ of sign vectors defined as follows. By definition, $\mathcal{F}^*(V)$ is the set of sign vectors $(\mathrm{sgn}(\lambda_1), \ldots, \mathrm{sgn}(\lambda_n)) \in \{-1, 0, +1\}^n$ given by non-trivial linear dependencies $\sum_{i=1}^n \lambda_i v_i = 0$ (with coefficients $\lambda_i \in \mathbb{R}$, not all of them zero). We call $\mathcal{F}^*(V)$ the dependency patterns of V. We define $\mathcal{F}(V)$ as the set of all sign vectors $(\mathrm{sgn}(\langle v_1, u \rangle), \ldots, \mathrm{sgn}(\langle v_n, u \rangle)) \in \{-1, 0, +1\}^n$, where u ranges over all non-zero vectors in $\mathbb{R}^3$ (equivalently, unit vectors in S^2). Thus, the elements $\mathcal{F}(V)$ encode partitions $V = V_- \sqcup V_0 \sqcup V_+$ by oriented planes through the origin, and we call them the dissection patterns of V. Equivalently, every vector $v_i \in V$ defines a great circle $H_i = \{x \in S^2 \mid \langle v_i, x \rangle = 0\}$ in S^2 and two open hemispheres $H_i^+ = \{x \in S^2 \mid \langle v_i, x \rangle > 0\}$ and $H_i^- = \{x \in S^2 \mid \langle v_i, x \rangle < 0\}$. The resulting arrangement $\mathcal{A}(V) = \{H_1^+, \ldots, H_n^+\}$ of hemispheres in S^2 determines a decomposition of S^2 into faces of dimensions $0, 1, 2$, where two points $u, u' \in S^2$ lie in the relative interior of the same face iff $\mathrm{sgn}(\langle v_i, u \rangle) = \mathrm{sgn}(\langle v_i, u' \rangle)$ for $1 \leqslant i \leqslant n$. Thus, we can identify each face of $\mathcal{A}(V)$ with its signature $F \in \mathcal{F}(V)$; by general position, the face with signature F has dimension $2 - |F_0|$ (there are no faces with $|F_0| > 2$, i.e., the arrangement is simple). Moreover, we call $|F_-|$ the level of the face. We refer to the correspondence between V and $\mathcal{A}(V)$ as polar duality (to distinguish it from Gale duality, see Sec. 3 below). Both $\mathcal{F}(V)$ and $\mathcal{F}^*(V)$ are invariant under invertible linear transformations of $\mathbb{R}^r$ and under positive rescaling (multiplying each vector v_i by some positive scalar $\alpha_i > 0$).

Definition 1 (f and f^*)**.** For integers s and t, define[1]

$$f_{s,t}(V) := |\{F \in \mathcal{F}(V) \mid |F_0| = s, |F_-| = t\}|, \qquad f^*_{s,t}(V) := |\{F \in \mathcal{F}^*(V) \mid |F_-| = t, |F_+| = s - t\}|$$

[1] By general position, $f_{s,t}(V) = 0$ unless $0 \leqslant s \leqslant d$ and $0 \leqslant t \leqslant n - s$, and $f^*_{s,t}(V) = 0$ unless $4 \leqslant s \leqslant n$ and $0 \leqslant t \leqslant s$. However, it will occasionally be convenient to allow an unrestricted range of indices.

Thus, $f_{s,t}(V)$ counts the $(2-s)$-dimensional faces of level t in $\mathcal{A}(V)$, and $f^*_{s,t}(V)$ counts the dependency patterns with t negative entries -1 and s non-zero entries ± 1. Together, these numbers form the f-matrix $f(V) = [f_{s,t}(V)]$ and the f^*-matrix $f^*(V) = [f^*_{s,t}(V)]$. Equivalently, we can encode this data into two bivariate polynomials $f_V(x,y)$ and $f^*_V(x,y)$ in $\mathbb{Z}[x,y]$, the f-polynomial and the f^*-polynomial of V, which are defined by

$$f_V(x,y) := \sum_{F \in \mathcal{F}(V)} x^{|F_0|} y^{|F_-|} = \sum_{s,t} f_{s,t}(V)\, x^s y^t,$$
$$f^*_V(x,y) := \sum_{F \in \mathcal{F}^*(V)} x^{|F_0|} y^{|F_-|} = \sum_{s,t} f^*_{s,t}(V)\, x^{n-s} y^t.$$

We are ready to state our first result. We say that $V = \{v_1, \ldots, v_n\} \subset \mathbb{R}^3$ is coneighborly if every open linear halfspace (bounded by a plane through the orgin) in $\mathbb{R}^3$ contains at least $\lfloor (n-2)/2 \rfloor$ of the vectors in V. Coneighborly configurations in $\mathbb{R}^3$ are precisely the Gale duals of $(n-4)$-dimensional neighborly polytopes with n vertices (see Section 3, and Example 8 for a specific example, cocyclic configurations). It is known [30] that there are at least $n^{\frac{3}{2}(1-o(1))n}$ combinatorially distinct $(n-4)$-dimensional neighborly polytopes with n vertices, hence at least that many different combinatorial types of coneighborly configurations of n vectors in S^2.

Theorem 2. Let $V = \{v_1, \ldots, v_n\} \subset \mathbb{R}^3$ be a vector configuration in general position. Then, for all $s \leqslant n$, the numbers $f^*_{s,0}(V)$ and $f^*_{s,\leqslant 1}(V) := f^*_{s,0}(V) + f^*_{s,1}(V)$ are maximized if V is coneighborly.

The result for $f^*_{s,0}(V)$ (and its generalization to vector configurations of arbitrary rank r) is known: It is equivalent, by Gale duality, to McMullen's Upper Bound Theorem for polytopes [27]; a direct proof was given by Welzl [40]. The result for $f^*_{s,\leqslant 1}(V)$ is new and confirms, for $d = n-4$, a generalization of the Upper Bound Theorem for sublevels of arrangements in S^d conjectured by Eckhoff [18], Linhart [22], and Welzl [40] (see Sec. 3, Conjecture 20).

1.2 The Spherical Arc Crossing Number of K_n

As an application of Theorem 2, we obtain a result about crossing numbers. Determining the crossing number $\mathrm{cr}(K_n)$ of the com-

plete graph K_n (the minimum number of crossings in any drawing of K_n in the plane $\mathbb{R}^2$, or equivalently in the sphere S^2, with edges represented by arbitrary Jordan arcs) is one of the foundational unsolved problems in geometric graph theory. This problem was first studied by Hill in the 1950's, who conjectured the following:

Conjecture 3 (Hill)**.**

$$\mathrm{cr}(K_n) = X(n) := \frac{1}{4}\left\lfloor \frac{n}{2} \right\rfloor \left\lfloor \frac{n-1}{2} \right\rfloor \left\lfloor \frac{n-2}{2} \right\rfloor \left\lfloor \frac{n-3}{2} \right\rfloor = \frac{3}{8}\binom{n}{4} + O(n^3) \quad (1)$$

This is known to hold for $n \leqslant 12$, but remains open in general (see [33, Sec. 1.3] or [36] for further background and references). There are several families of drawings showing that $\mathrm{cr}(K_n) \leqslant X(n)$ for all n, but the best lower bound to date is $\mathrm{cr}(K_n) \geqslant 0.985 \cdot X(n)$ [14]. Here, we prove Hill's conjecture for the following class of drawings. Let $V = \{v_1, \ldots, v_n\} \subset S^2$ be a configuration of n unit vectors in general position. If we connect every pair of vectors in V by the shortest geodesic arc between them in S^2 (which is unique, since no two vectors are antipodal, by general position) we obtain a drawing of the complete graph K_n in S^2, which we call a spherical arc drawing. Let $\mathrm{cr}(V)$ denote the number of crossings in this drawing.

Theorem 4. For every configuration of n vectors in general position in S^2,

$$\mathrm{cr}(V) \geqslant \frac{1}{4}\left\lfloor \frac{n}{2} \right\rfloor \left\lfloor \frac{n-1}{2} \right\rfloor \left\lfloor \frac{n-2}{2} \right\rfloor \left\lfloor \frac{n-3}{2} \right\rfloor$$

Moreover, the lower bound is attained with equality if V is coneighborly.

The fact that coneighborly configurations yield spherical arc drawings of K_n achieving the number $X(n)$ of crossings in Hill's conjecture and the connection to the Eckhoff–Linhart–Welzl conjecture were first observed by Wagner [38, 39]. For a connection to geometric probability, in particular to old results of Wendel [41] and Moon [28], see Remark 16. Spherical arc drawings generalize the well-studied class of rectilinear drawings of K_n, given by n points in general position in $\mathbb{R}^2$ connected by straight-line segments; more precisely, rectilinear drawings correspond to the sub-class of sperical arc drawings for which the vector

configuration V is pointed, i.e., contained in an open linear halfspace. Pointed configurations $V \subset \mathbb{R}^3$ correspond to point sets $S \subset \mathbb{R}^2$ by radial projection,[2] and under this correspondence, $\mathcal{F}(V)$ and $\mathcal{F}^*(V)$ encode the partitions of S by affine lines and Radon partitions $S = S_- \sqcup S_0 \sqcup S_+$, $\operatorname{conv}(S_+) \cap \operatorname{conv}(S_-) \neq \emptyset$, respectively. Moreover, for a pointed $V \subset S^2$, spherical arcs correspond to straight-line segments. Theorem 4 complements earlier results of Lovász, Vesztergombi, Wagner, and Welzl [24] and Ábrego and Fernández-Merchant [6], who showed that the rectilinear crossing number $\overline{\operatorname{cr}}(K_n)$ (the minimum number of crossings in any rectilinear drawing of K_n) is at least $X(n)$; in fact, $\overline{\operatorname{cr}}(K_n) \geqslant (\frac{3}{8} + \varepsilon + o(1))\binom{n}{4}$ for some constant $\varepsilon > 0$ [24]. Thus, unlike the spherical arc crossing number, the rectilinear crossing number $\overline{\operatorname{cr}}(K_n)$ is strictly larger than $X(n)$ (and hence larger than $\operatorname{cr}(K_n)$) in the asymptotically leading term. We refer to [8] for a detailed survey, including a series of subsequent improvements [15, 9, 7, 1] leading to the currently best bound [5] $\overline{\operatorname{cr}}(K_n) > 277/729\binom{n}{4} + O(n^3) > 0.37997\binom{n}{4} + O(n^3)$. We remark that the arguments in [24, 6] have been generalized to verify Hill's conjecture for other classes of drawings, including 2-page drawings [3], monotone drawings [13], cylindrical, x-bounded, and shellable drawings [4], bishellable drawings [2], and seq-shellable drawings [29]. Currently we do not know how spherical arc drawings relate to these other classes of drawings.

1.3 The g-Matrix

The central new notion of this paper is the g-matrix $g(V \to W)$ of a pair of vector configurations, which encodes the differences $f(W) - f(V)$ and $f^*(W) - f^*(V)$ of f-matrices and f^*-matrices and which generalizes both the classical g-vector of a simple polytope and a Gale dual version of the g-vector studied by Lee [21] and Welzl [40]. The definition of the g-matrix (which we present here for vector configurations in $\mathbb{R}^3$ and which easily generalizes to higher dimensions, see [35]) is based on how $\mathcal{F}$ and $\mathcal{F}^*$ (and hence f and f^*) change by mutations during a continuous motion

[2] Given $S = \{p_1, \ldots, p_n\} \subset \mathbb{R}^2$, we obtain a pointed vector configuration $V = \{v_1, \ldots, v_n\} \subset \mathbb{R}^3$ by setting $v_i := (1, p_i) \in \{1\} \times \mathbb{R}^d \subset \mathbb{R}^3$. Conversely, if $V \subset \{x \in \mathbb{R}^3 \colon \langle u, x\rangle > 0\}$ for some $u \in S^2$ then we radially project each vector v_i to the point $p_i := \frac{1}{\langle u, v_i\rangle} v_i$ in the tangent plane $\{x \in \mathbb{R}^r \colon \langle u, x\rangle = 1\} \cong \mathbb{R}^2$.

(this idea has a long history in discrete geometry, including in the study of dissection patterns, see, e.g., [11]). During a mutation, a unique triple of vectors, indexed by some $R = \{i_1, i_2, i_3\} \subset [n]$, become linearly dependent, the orientation of this triple changes, and all other triples remain linearly independent. Let V and W denote the vector configurations before and after the mutation. In terms of the polar dual arrangements in S^2, the three great circles indexed by R intersect in a pair of antipodal points $u, -u \in S^2$ during the mutation. Immediately before and immediately after the mutation, these three great circles bound an antipodal pair of small triangular faces $\sigma, -\sigma$ in $\mathcal{A}(V)$ and a corresponding pair $\tau, -\tau$ of triangular faces in $\mathcal{A}(W)$, respectively, see Figures 1 and 2. We say that σ and $-\sigma$ disappear during the mutation, and that τ and $-\tau$ appear. Let $Y \in \mathcal{F}(W)$ be the signature of τ. Set $j := |R \cap Y_-|$ and $k := |([n] \setminus R) \cap Y_-|$; we call (j, k) the type of τ. It is easy to see that σ has type $(3-j, k)$. Analogously, $-\tau$ and $-\sigma$ are of type $(3-j, n-3-k)$ and $(j, n-3-k)$, respectively. Any two configurations $V = \{v_1, \dots, v_n\}$ and $W = \{w_1, \dots, w_n\}$ of n vectors in general position in $\mathbb{R}^3$ are connected by a finite sequence of mutations. We define $g_{j,k}(V \to W)$ as the net number of triangular faces of type (j, k) that appear during this sequence (every mutation contributes $+1$, -1, or 0), see Definition 24 for the details; this depends only on V and W, not on the sequence of mutations, see Remark 7. Analyzing how the f-polynomial and the f^*-polynomial change during a mutation (Lemmas 22 and 23) yields the following:

Theorem 5. Let $V, W \in \mathbb{R}^{3\times n}$ be a pair of vector configurations in general position. The g-matrix $g(V \to W)$ of the pair is an $4\times(n-4)$-matrix with integer entries $g_{j,k} := g_{j,k}(V \to W)$, $0 \leqslant j \leqslant 3$, $0 \leqslant k \leqslant n-3$, which has the following properties:

1. $g(W \to V) = -g(V \to W)$. If $U \in \mathbb{R}^{r\times n}$, then $g(U \to W) = g(U \to V) + g(V \to W)$.

2. For $0 \leqslant j \leqslant 3$ and $0 \leqslant k \leqslant n-3$, the g-matrix satifies the skew-symmetries

$$g_{j,k} = -g_{3-j,k} = -g_{j,n-3-k} = g_{3-j,n-3-k} \tag{2}$$

   Thus, the g-matrix is determined by the submatrix $[g_{j,k} \colon 0 \leqslant j \leqslant 1, 0 \leqslant k \leqslant \lfloor \frac{n-4}{2} \rfloor]$, which we call the small g-matrix. Equivalently, the g-polynomial $g(x, y) := g_{V\to W}(x, y) := \sum_{j,k} g_{j,k} x^j y^k \in$

$\mathbb{Z}[x,y]$ satisfies

$$g(x,y) = -x^3 g(\tfrac{1}{x}, y) = -y^{n-3} g(x, \tfrac{1}{y}) = x^3 y^{n-3} g(\tfrac{1}{x}, \tfrac{1}{y}) \qquad (3)$$

3. The g-polynomial determines the difference $f(x,y) := f_W(x,y) - f_V(x,y)$ by

$$f(x,y) = (1+x)^3 g(\tfrac{x+y}{1+x}, y) = \textstyle\sum_{j=0}^{3} \sum_{k=0}^{n-3} g_{j,k} \cdot (x+y)^j (1+x)^{3-j} y^k \quad (4)$$

4. The g-polynomial determines the difference $f^*(x,y) = f^*_W(x,y) - f^*_V(x,y)$ by

$$f^*(x,y) = \textstyle\sum_{j,k} \underbrace{g_{j,k}(W \to V)}_{=-g_{j,k}(V \to W)} (x+y)^k (x+1)^{n-3-k} y^j = -(x+1)^{n-3} g(y, \tfrac{x+y}{x+1}) \qquad (5)$$

By comparing coefficients in (4) and (5), we get:

Corollary 6. For $0 \leqslant s \leqslant 2$ and $0 \leqslant t \leqslant n$,

$$f_{s,t}(W) - f_{s,t}(V) = \sum_{j,k} \binom{j}{t-k} \binom{3-j}{s-j+t-k} g_{j,k}(V \to W), \qquad (6)$$

Moreover, for $4 \leqslant s \leqslant n$ and $0 \leqslant t \leqslant s$,

$$f^*_{s,t}(V) - f^*_{s,t}(W) = \sum_{j,k} \binom{k}{t-j} \binom{n-3-k}{s-t+j-3} g_{j,k}(V \to W) \qquad (7)$$

In particular (by specializing (7) to $t = 0, 1$), we get that for $4 \leqslant s \leqslant n$

$$f^*_{s,0}(V) - f^*_{s,0}(W) = \sum_{k} \binom{n-3-k}{s-3} g_{0,k}(V \to W) \qquad (8)$$

and

$$f^*_{1,0}(V) - f^*_{1,0}(W) = \textstyle\sum_k \binom{n-3-k}{s-3} g_{1,k}(V \to W) + \sum_k k \binom{n-3-k}{s-4} g_{0,k}(V \to W) \qquad (9)$$

Remark 7. It is easy to show that the system of equations (8) can be inverted, hence the differences $f^*_{s,0}(V) - f^*_{s,0}(W)$, $4 \leqslant s \leqslant n$, determine $g_{0,k}(V \to W)$, $0 \leqslant k \leqslant n-3$; by the same token, the numbers $g_{1,k}(V \to W)$ are then determined by $f^*_{1,0}(V) - f^*_{1,0}(W)$,

$4 \leqslant s \leqslant n$, together with the already determined $g_{0,k}(V \to W)$, $0 \leqslant k \leqslant n-3$. By skew-symmetry, this determines the entire g-matrix $g(V \to W)$, which thus only depends on the difference $f^*(V) - f^*(W)$ of f^*-matrices. Analogously, the g-matrix is determined by the difference of f-matrices. Thus, Theorem 5 could be taken as a formal definition of the g-matrix. Moreover, the skew-symmetry $g(x,y) = -x^r g(\frac{1}{x}, y)$ reflects the Dehn–Sommerville relation (21), and the symmetry $g(x,y) = x^r y^{n-r} g(\frac{1}{x}, \frac{1}{y})$ reflects the antipodal symmetry (20).

We will need the following two special configurations:

Example 8 (Cyclic and Cocyclic Configurations)**.** Let $t_1 < t_2 < \cdots < t_n$ be real numbers and define $v_i := (1, t_i, t_i^2) \in \mathbb{R}^3$. We call $V_{\text{cyclic}}(n,3) := \{v_1, \dots v_n\}$ and $V_{\text{cocyclic}}(n,3) := \{(-1)^i v_i \colon 1 \leqslant i \leqslant n\}$ the cyclic and cocyclic configurations of n vectors in $\mathbb{R}^3$. (the combinatorial types of these configurations are independent of the choice of the t_i). The configuration $V_{\text{cyclic}}(n,3)$ is pointed and corresponds to a point set in $\mathbb{R}^2$ in convex position (points on a parabola); it follows that $f^*_{s,0}(V_{\text{cyclic}}(n,3)) = f^*_{s,1}(V_{\text{cyclic}}(n,3)) = 0$ for all s. Moreover, it is easy to show [26, Lemma 3.5.1] that $V_{\text{cocyclic}}(n,3)$ is coneighborly.

We are now ready to define the g-matrix and the g^*-matrix of a vector configuration.

Definition 9 (g-matrix and g^*-matrix)**.** Let V be a configuration of n vectors in $\mathbb{R}^3$. Set

$$g_{j,k}(V) := g_{j,k}(V_{\text{cocyclic}}(n,3) \to V), \qquad \text{and} \qquad g^*_{j,k}(V) := g_{j,k}(V \to V_{\text{cyclic}}(n,3))$$

for $0 \leqslant j \leqslant r$ and $0 \leqslant k \leqslant n-r$. We call $g(V) = [g_{j,k}(V)]$ and $g^*(V) = [g^*_{j,k}(V)]$ the g-matrix and the g^*-matrix of V, respectively.

Using skew-symmetry, Cor. 6, and the properties of cyclic configurations, we get

Lemma 10. Let $V \in \mathbb{R}^{3\times n}$ be a vector configuration in general position and let $s \geqslant 4$.

$$f^*_{s,0}(V) = \sum_{k=0}^{\lfloor \frac{n-4}{2} \rfloor} \underbrace{\left(\binom{n-3-k}{s-3} - \binom{k}{s-3} \right)}_{\geqslant 0} g^*_{0,k}(V)$$

and

$$f^*_{s,\leqslant 1}(V) = \sum_{k=0}^{\lfloor \frac{n-4}{2} \rfloor} \underbrace{\left(\binom{n-3-k}{s-3} - \binom{k}{s-3} \right)}_{\geqslant 0} g^*_{1,k}(V) + \sum_{k=0}^{\lfloor \frac{n-4}{2} \rfloor} \underbrace{\left(\binom{n-3-k}{s-4} k - \binom{k}{s-4} (n-3-k) \right)}_{\geqslant 0} g^*_{0,k}(V)$$

The numbers $g^*_{0,k}$, $0 \leqslant k \leqslant n-3$, correspond to a Gale dual version of the g-vector of convex polytopes studied by Lee [21] and Welzl [40]. The following is implicit in [40, Sec. 4]:

Theorem 11. $V \subset \mathbb{R}^3$ be a configuration of n vectors in general position. Then

$$g^*_{0,k}(V) \leqslant \binom{k+2}{2} \qquad (0 \leqslant k \leqslant \lfloor \frac{n-4}{2} \rfloor)$$

Equality holds if V is coneighborly.

To prove Theorem 2 (from which Theorem 4 follows), we need two additional new results.

Lemma 12. Let $V \subset \mathbb{R}^3$ be a configuration of n vectors in general position. Then

$$g_{1,k}(V) + g^*_{1,k}(V) = (k+1)n - 3\binom{k+2}{2} \qquad (0 \leqslant k \leqslant \lfloor \frac{n-4}{2} \rfloor)$$

Our main result is the following:

Theorem 13. Let $V \subset \mathbb{R}^3$ be a configuration of n vectors in general position. Then $g_{1,k}(V) \geqslant 0$ for $0 \leqslant k \leqslant \lfloor \frac{n-4}{2} \rfloor$; equivalently (by Lemma 12),

$$g^*_{1,k}(V) \leqslant (k+1)n - 3\binom{k+2}{2} \qquad (0 \leqslant k \leqslant \lfloor \frac{n-4}{2} \rfloor)$$

Equality holds if V is coneighborly.

Proof of Theorem 2. By Lemma 10, for $s \geqslant 4$, $f^*_{s,0}(V)$ and $f^*_{s,\leqslant 1}(V)$ are non-negative linear combinations of the numbers $g^*_{0,k}(V)$ and $g^*_{1,k}(V)$, $0 \leqslant k \leqslant \lfloor \frac{n-4}{2} \rfloor$. Hence, Theorem 2 follows from Theorems 11 and 13. □

Remark 14. The definition of the g-matrix generalizes to vector configurations $V \subset \mathbb{R}^r$, for arbitrary $r = d + 1 \geqslant 1$, see the paper [35] by Streltsova and Wagner, who use this to determine all linear relations between face numbers of levels in arrangements in S^d. We conjecture that the small g-matrix of any vector configuration in $\mathbb{R}^r$ is nonnegative. This would generalize the Generalized Lower Bound Theorem for polytopes and imply the Eckhoff–Linhart–Welzl conjecture in full generality.

2 Bounding the Spherical Arc Crossing Number

Observation 15. Let $V \subset \mathbb{R}^3$ be a configuration of n vectors in general position. Then

$$f^*_{4,0}(V) + f^*_{4,1}(V) + \frac{1}{2} f^*_{4,2}(V) = \binom{n}{4}$$

If all vectors in V have unit length (which we can achieve by positive rescaling) then the number of crossings in the induced spherical arc drawing of K_n equals $\mathrm{cr}(V) = \frac{1}{2} f^*_{4,2}(V)$.

Proof. By general position, for every quadruple $W = \{w_1, w_2, w_3, w_4\} \subset V$, the space of linear dependencies of W has dimension 1, and hence $\mathcal{F}^*(W)$ consists of exactly two sign vectors $\{F, -F\}$. Thus, there are exactly three combinatorial types of quadruples (see Fig. 3):

Type 0: $f^*_{4,0}(W) = 1$ (geometrically, W forms the vertex set of a tetrahedron containing the origin in its interior);

Type 1: $f^*_{4,1}(W) = 1$ (W spans a pointed cone with three extremal rays that contains the fourth vector in its interior);

Type 2: $f^*_{4,2}(W) = 2$ (W spans a pointed cone with four extremal rays).

Every quadruple of Type 2 contributes exactly one crossing, other types do not. □

Proof of Theorem 4. By Observation 15, Theorem 4 is equivalent to the statement that $f^*_{4,2}(V) \geqslant 2X(n)$, equivalently, $f^*_{4,0}(V) +$

$f^*_{4,1}(V) \leqslant \binom{n}{4} - X(n)$, where equality holds in both bounds if V is coneighborly. By the special case $s = 4$ of Theorem 2,

$$f^*_{4,\leqslant 1}(V) \leqslant \sum_{k=0}^{\lfloor \frac{n-4}{2} \rfloor} (n-3-2k)\left((k+1)n - 3\binom{k+2}{2}\right) =: Y(n), \qquad (10)$$

with equality if V is coneighborly. A straightforward calculation (see Appendix A) shows that $Y(n) = \binom{n}{4} - X(n)$. □

Remark 16. We mention a connection to geometric probability. Let μ be a probability distribution on $\mathbb{R}^3$; we assume that μ is non-degenerate in the sense that every plane through the origin has μ-measure zero. A beautiful argument due to Wendel [41] shows that if μ is centrally symmetric and $W = \{w_1, w_2, w_3, w_4\} \subset \mathbb{R}^3$ is a set of four independent μ-random vectors then the probability that W is of Type $i \in \{0, 1, 2\}$ equals q_i, where $q_0 = \frac{\binom{4}{0}+\binom{4}{4}}{2^4} = \frac{1}{8}$, $q_1 = \frac{\binom{4}{1}+\binom{4}{3}}{2^4} = \frac{1}{2}$, and $q_2 = \frac{\binom{4}{2}}{2^4} = \frac{3}{8}$; it follows that for any set $V \subset \mathbb{R}^3$ of n independent μ-random vectors, the expected number of quadruples of type i equals $q_i\binom{n}{4}$. In particular, if the vectors in V are chosen independently and uniformly at random from S^2 then the expected number of crossings in the spherical arc drawing given by V is $\frac{3}{8}\binom{n}{4}$, which was also independently shown by Moon [28]. Theorem 4, together with a well-known limiting argument [34] implies that for an arbitrary (not necessarily centrally symmetric) non-degenerate probability distribution μ on S^2, the expected number of crossings in the spherical arc drawing given by n independent μ-random points is at least $\frac{3}{8}\binom{n}{4}$.

3 Gale Duality and Exact Upper Bounds for Sublevels

The definitions of the dissection patterns $\mathcal{F}(V)$, the dependency patterns $\mathcal{F}^*(V)$, the matrices $f(V)$ and $f^*(V)$, and the polynomials $f_V(x, y)$ and $f^*_V(x, y)$ readily generalize to vector configurations $V = \{v_1, \ldots, v_n\} \subset \mathbb{R}^r$ in general position and the polar dual spherical arrangements $\mathcal{A}(V)$ in S^d, for $r \geqslant 1$ and $d = r - 1$. In general, $f_{s,t}(V)$ counts the $(d-s)$-faces of level t in the simple arrangement $\mathcal{A}(V)$, and $f^*_{s,t}(V)$ counts the number of linear dependency sign patterns

with t negative entries -1 and s non-zero entries ± 1. It is well-known that $\mathcal{F}(V)$ and $\mathcal{F}^*(V)$ determine each other; moreover, the polynomials $f_V(x,y)$ and $f_V^*(x,y)$ also determine each other [35, Theorem 23] via the identity

$$f_V^*(x,y) = (x+y+1)^n - (-1)^r x^n - (x+1)^n f_V(-\tfrac{x}{x+1}, \tfrac{x+y}{x+1}) \qquad (11)$$

Definition 17 (Gale Duality)**.** Two vector configurations $V \in \mathbb{R}^{r\times n}$ and $W \in \mathbb{R}^{(n-r)\times n}$ are called Gale duals of one another if the rows of V and the rows of W span subspaces of $\mathbb{R}^n$ that are orthogonal complements of one another. Since we always assume that V and W are in general position and of full rank, this is equivalent to the condition $VW^\top = 0$.

Gale dual configurations determine each other up to linear isomorphisms of their ambient spaces $\mathbb{R}^r$ and $\mathbb{R}^{n-r}$, respectively. Thus, we speak of the Gale dual of V, which we denote by V^*. Obviously, $(V^*)^* = V$. Moreover, it follows straightforwardly from the definition that

$$\mathcal{F}^*(V) = \mathcal{F}(V^*), \qquad \text{hence} \quad f_V^*(x,y) = f_{V^*}(x,y) \quad \text{and} \quad f_{s,t}^*(V) = f_{n-s,t}(V^*)$$

The construction from Example 8 generalizes to yield cyclic configurations $V_{\text{cyclic}}(n,r)$ and cocyclic configurations $V_{\text{cocyclic}}(n,r)$ for arbitrary $n \geqslant r \geqslant 1$ [42, 26]. These are examples of neighborly and coneighborly configurations, which we define next. Let $V = \{v_1, \ldots, v_n\} \subseteq \mathbb{R}^r$ be a vector configuration in general position. A subset $W \subseteq V$ is extremal if there exists an oriented linear hyperplane H that contains all vectors in W and such that one of the two open halfspaces bounded by H contains $V \setminus W$.

Definition 18 (Neighborly and Coneighborly Configurations)**.** A vector configuration $V = \{v_1, \ldots, v_n\} \subseteq \mathbb{R}^r$ is coneighborly if every open linear halfspace contains at least $\lfloor \frac{n-r+1}{2} \rfloor$ vectors of V. It is neighborly if every subset $W \subseteq V$ of size $|W| \leqslant \lfloor \frac{r-1}{2} \rfloor$ is extremal.

Every neighborly vector configuration $V \subset \mathbb{R}^r$ is pointed, hence corresponds to a point set $S \subset \mathbb{R}^d$, $d = r-1$, and V being neighborly means the simplicial d-polytope $\operatorname{conv}(S)$ is a neighborly polytope, i.e., every subset of S of size at most $\lfloor \frac{d}{2} \rfloor$ forms a face. We note that for $r = 1, 2$ ($d = 0, 1$) neighborliness is the same as being pointed, and for $r = 3, 4$ ($d = 2, 3$) V is neighborly iff the point set S is in convex position. By a celebrated result of McMullen [27] neighborly polytopes maximize the number of faces of any dimension:

Theorem 19 (Upper Bound Theorem for Convex Polytopes)**.** Let $V \subset \mathbb{R}^r$ be a configuration of n vectors in general position. Then

$$f_{s,0}(V) \leqslant f_{s,0}(V_{\text{cyclic}}(n,r)) \qquad (0 \leqslant s \leqslant d = r-1)$$

with equality if V is neighborly.

Eckhoff [18, Conj. 9.8], Linhart [22], and Welzl [40], independently of one another (and in slightly different forms) conjectured a far-reaching generalization of Theorem 19 for sublevels of arrangements. Let $f_{s,\leqslant k}(V) := \sum_{t \leqslant k} f_{s,t}(V)$.

Conjecture 20 (Generalized Upper Bound Conjecture for Sublevels)**.** Let $V \subset \mathbb{R}^r$ be a configuration of n vectors in general position, and $0 \leqslant k \leqslant \lfloor \frac{n-r-1}{2} \rfloor$. Then

$$f_{s,\leqslant k}(V) \leqslant f_{s,\leqslant k}(V_{\text{cyclic}}(n,r)) \qquad (0 \leqslant s \leqslant d = r-1)$$

with equality if V is neighborly.

A random sampling argument due to Clarkson (see [16]) shows that Conjecture 20 is true asymptotically, for fixed r and $n, k \to \infty$, up to a constant factor depending on r. For the case $s = d$ of vertices at sublevel $(\leqslant k)$, Conjecture 20 was shown to be true exactly (without constant factors) by Peck [31] and by Alon and Győri [10] for $r \leqslant 3$ and by Welzl [40] for pointed vector configurations in rank $r = 4$. Wagner [38] proved that it is true up to a factor of 4 for arbitrary rank r. Explicit formulas for the numbers $f_{s,t}(V_{\text{cyclic}}(n,r))$ have been obtained by Andrzejak and Welzl [12]. By Gale duality, Theorem 2 confirms Conjecture 20 for $n = r + 3 = d + 4$.

Remark 21. Bounding the maximum number of faces at level exactly k is of a rather different flavor. For coneighborly configurations, all $2\binom{n}{r-1}$ vertices of the dual arrangement are concentrated at one or two consecutive levels $k = \lfloor \frac{n-r+1}{2} \rfloor$ and $k = \lceil \frac{n-r+1}{2} \rceil$. By contrast, determining the maximum number $f_{d,k}$ of vertices at level k for pointed vector configurations in $\mathbb{R}^r$ is a difficult open problem, first studied by Lovász [23] and Erdős, Lovász, Simmons, and Straus [20] in the 1970s (see [39] or [25, Ch. 11] for more details and background); even for $r = 3$ (i.e., $d = 2$) there remains a big gap between the best upper and lower bounds to date, which are $O(nk^{1/3})$ (due to Dey [17]) and $ne^{\Omega(\sqrt{\log k})}$ (due to Tóth [37]), respectively.

4 Continuous Motion and the g-Matrix

Any two configurations $V = \{v_1, \ldots, v_n\}$ and $W = \{w_1, \ldots, w_n\}$ of n vectors in general position in $\mathbb{R}^3$ can be deformed into one another through a continuous family $V(t) = \{v_1(t), \ldots, v_n(t)\}$ of vector configurations, where $v_i(t)$ describes a continuous path from $v_i(0) = v_i$ to $v_i(1) = w_i$ in $\mathbb{R}^3$. If we choose this continuous motion sufficiently generically, then there be a finite set of events $0 < t_1 < \cdots < t_N < 1$ during which the combinatorial type of $V(t)$ (encoded by $\mathcal{F}(V(t))$) changes in a controlled way, by a mutation, as described in Section 1.3. Thus, any two vector configurations are connected by a finite sequence $V = V_0, V_1, \ldots, V_N = W$ such that V_{s-1} and V_s differ by a mutation, $1 \leqslant s \leqslant N$. We describe the change from $\mathcal{F}(V)$ to $\mathcal{F}(W)$ when V and W differ by a single mutation. Let $R \in \binom{[n]}{3}$ index the unique triple of vectors that become linearly dependent during the mutation; the corresponding triple of great circles in S^2 intersect in a pair of antipodal points during the mutation. As discussed above, immediately before and immediately after the mutation, these three great circles bound an antipodal pair of triangular faces $\sigma, -\sigma$ in $\mathcal{A}(V)$ (the disappearing triangles) and a corresponding pair of triangular faces $\tau, -\tau$ in $\mathcal{A}(W)$ (the appearing triangles), respectively, see Figures 1 and 2. Recall that the type of τ is defined as the pair (j, k), where $j = |R \cap Y_-|$ and $k = |([n] \setminus R) \cap Y_-|$ and $Y \in \mathcal{F}(W)$ is the signature of τ, The types of σ, $-\tau$ and $-\sigma$ are $(3-j, k)$, $(3-j, n-3-k)$ and $(j, n-3-k)$, respectively, see Figures 1 and 2. We say that the mutation $V \to W$ is of type $(j, k) \equiv (3-j, n-3-k)$ (the pair of types of the triangles $\tau, -\tau$ that appear). We have $F \in \mathcal{F}(V) \setminus \mathcal{F}(W)$ iff F corresponds to a face of $\mathcal{A}(V)$ contained in one of the disappearing triangular cells $\sigma, -\sigma$, and $F \in \mathcal{F}(W) \setminus \mathcal{F}(V)$ iff F corresponds to a face of $\mathcal{A}(W)$ contained in one of the appearing triangular cells $\tau, -\tau$. All other faces are preserved, i.e., they belong to $\mathcal{F}(V) \cap \mathcal{F}(W)$. Let us define $f_\sigma(x, y) := \sum_{F \subseteq \sigma} x^{|F_0|} y^{|F_-|}$, where we use the notation $F \subseteq \sigma$ to indicate that the sum ranges over all $F \in \mathcal{F}(V)$ corresponding to faces of $\mathcal{A}(V)$ contained in σ. The polynomials $f_{-\sigma}(x, y)$, $f_\tau(x, y)$, and $f_{-\tau}(x, y)$ are defined analogously. These four polynomials have a simple form:

$$f_\sigma(x,y) = y^k \left[(x+1)^j (x+y)^{3-j} - x^3\right], \quad f_{-\sigma}(x,y) = y^{n-3-k} \left[(x+1)^{3-j}(x+y)^j - x^3\right]$$
$$f_\tau(x,y) = y^k \left[(x+1)^{3-j} (x+y)^j - x^3\right], \quad f_{-\tau}(x,y) = y^{n-3-k} \left[(x+1)^j (x+y)^{3-j} - x^r\right]$$

Thus, we get the following:

Lemma 22. Let $V \to W$ be a mutation of Type $(j,k) \equiv (3-j, n-3-k)$. Then

$$f_W(x,y) - f_V(x,y) = \left(y^k - y^{n-3-k}\right)\left[(x+1)^{3-j}(x+y)^j - (x+1)^j(x+y)^{3-j}\right] \tag{12}$$

As remarked in Sec. 3 above, the polynomials $f_V(x,y)$ and $f_V^*(x,y)$ determine each other via the identity (11). This implies the following (which can also be proven by a direct analysis of how $\mathcal{F}^*$ and hence f^* change during a mutation):

Lemma 23. Let $V \to W$ be a mutation of Type $(j,k) \equiv (3-j, n-3-k)$. Then

$$f_W^*(x,y) - f_V^*(x,y) = \left(y^j - y^{3-j}\right)\left[(x+1)^k(x+y)^{n-3-k} - (x+1)^{n-3-k}(x+y)^k\right] \tag{13}$$

We are now ready to define the g-matrix $g(V \to W)$ of a pair of vector configurations.

Definition 24 (g-Matrix of a pair)**.** Let V, W be configurations of n vectors in $\mathbb{R}^3$. If $V \to W$ is a single mutation of Type $(i,\ell) \equiv (r-i, n-3-\ell)$ then we define the g-matrix $g(V \to W) = [g_{j,k}(V \to W)]$, $0 \leqslant j \leqslant 3$ and $0 \leqslant k \leqslant n-3$, as follows: If $2\ell = n-3$, then $g_{j,k}(V \to W) = 0$ for all j,k. If $2\ell \neq n-3$, then

$$g_{j,k}(V \to W) := \begin{cases} +1 & \text{if } (j,k) = (i,\ell) \text{ or } (j,k) = (3-i, n-3-\ell) \\ -1 & \text{if } (j,k) = (3-i,\ell) \text{ or } (j,k) = (i, n-3-\ell) \\ 0 & \text{else.} \end{cases}$$

More generally, if V and W are connected by a sequence $V = V_0, V_1, \ldots, V_N = W$, where V_{s-1} and V_s differ by a single mutation, then we define

$$g_{j,k}(V \to W) := \sum_{s=1}^{N} g_{j,k}(V_{s-1} \to V_s)$$

Proof of Theorem 5. The skew-symmetries of the g-matrix and the fact that $g(V \to W) = -g(W \to V)$ and $g(U \to W) = g(U \to V) + g(V \to W)$ follow immediately from the definition. Moreover, (4) and (5) follow from Lemmas 22 and 23. □

By Gale duality, Theorem 5 implies:

Corollary 25. Let $V, W \in \mathbb{R}^{r\times n}$ be vector configurations, and let $V^*, W^* \in \mathbb{R}^{(n-r)\times n}$ be their Gale duals. Then

$$g_{j,k}(V \to W) = -g_{k,j}(V^* \to W^*).$$

5 The Upper Bound for $g^*_{1,k}$ in Rank 3

In this section, we prove Lemma 12 and Theorem 13. Let $V = \{v_1, \ldots, v_n\} \subset \mathbb{R}^3$ be a configuration of n vectors in general position. It will be convenient to view V as the set $S - o = \{v_i = p_i - o \colon 1 \leqslant i \leqslant n\}$ of differences between a point set $S = \{p_1, \ldots, p_n\} \subset \mathbb{R}^3$ (the tips of the vectors) and a fixed point $o \in \mathbb{R}^3$ (the origin). Choose a line ℓ in $\mathbb{R}^3$ through o in general position (in particular, ℓ is not contained in the linear span of any pair of vectors in V). By positively rescaling the vectors v_i, we may assume that S is a subset of the cylinder Z consisting of the points in $\mathbb{R}^3$ at Euclidean distance exactly 1 from ℓ. It follows that S is a point set in convex position. In what follows, we will consider vector configurations of the form $S - p_0$, where $p_0 \in \ell$, and how these configurations change as we move p_0 continuously along ℓ. Given $S = \{p_1, \ldots, p_n\}$ and $p_0 \in \ell$, define $w_i := (1, p_i) \in \{1\} \times \mathbb{R}^3 \subset \mathbb{R}^4$, $u_i := \frac{1}{\|w_i\|} w_i \in S^3$, $0 \leqslant i \leqslant n$, and $U := \{u_1, \ldots, u_n\}$. Let U/u_0 be the orthogonal projection of U onto the hyperplane $u_0^\perp$ (viewed as a vector configuration in $u_0^\perp \cong \mathbb{R}^3$). Note that U/u_0 has the same combinatorial type as $S - p_0$. In particular, for $p_0 = o$, U/u_0 has the same combinatorial type as $V = S - o$. As p_0 moves along ℓ, the corresponding vector $u_0 \in S^3$ moves along an open great semi-circle in S^3 (the radial projection of the line $\{1\} \times \ell$ onto S^3), whose closure is a closed great semi-circle γ in S^3 that connects two antipodal points $u_{-\infty}$ and $u_{+\infty}$ (corresponding to "points on ℓ at $\pm\infty$"). By general position of ℓ, γ is in general position with respect to U.

Lemma 26. The pair consisting of the vector configuration $U = \{u_1, \ldots, u_n\}$ and the semi-circle $\gamma \subset S^3$ has the following properties:

1. The rank 4 vector configuration $U \subset S^3 \subset \mathbb{R}^4$ is neighborly.
2. The rank 3 vector configuration $U/u_{-\infty} = U/u_{+\infty}$ is neighborly.

3. Let $u_0 \in \gamma \subset S^3$ and $1 \leqslant i \leqslant n$. Then the vector u_i is extremal in $U \sqcup \{u_0\}$ (i.e., there exists $w \in S^3$ such that $\langle w, u_i \rangle = 0$ and $\langle w, u_s \rangle > 0$ for $s \in \{0\} \sqcup ([n] \setminus \{i\})$).

Proof. Since the point set $S \subset \mathbb{R}^3$ is in convex position (by virtue of being contained in the cylinder Z), the vector configuration $U := \{u_1, \ldots, u_n\} \subset S^3 \subset \mathbb{R}^4$ is neighborly. Moreover, linear hyperplanes in $\mathbb{R}^4$ orthogonal to $u_{\pm\infty}$ correspond to affine planes in $\mathbb{R}^3$ that are parallel to the line ℓ, or equivalently to affine lines in the plane $\ell^\perp$ orthogonal to ℓ. Thus, the vector configuration $U/u_{-\infty} = U/u_{+\infty}$ is neighborly because the orthogonal projection of S to $\ell^\perp \cong \mathbb{R}^2$ is in convex position. Finally, the third property follows because for every point $p_i \in S$, the tangent plane to the cylinder Z at p_i contains no other point of P, by general position, hence $\{p_0\} \sqcup P \setminus \{p_i\}$ is contained in one of the open affine halfspaces determined by that tangent plane. □

For $0 \leqslant i \leqslant n$, let $H_i := \{x \in S^3 \colon \langle u_i, x \rangle = 0\}$, and let $\mathcal{A}(U)$ be the arrangement of hemispheres $H_1^+, \ldots, H_n^+$ in S^3. Note that the rank 3 vector configuration U/u_0 is polar dual to the 2-dimensional arrangement $\mathcal{A}(U) \cap H_0$ in $H_0 \cong S^2$.

Definition 27 (k-Arcs, $\Lambda_k(U)$, $\lambda_k(U, u_0)$)**.** Let $k \leqslant \frac{n-4}{2}$. Let $C = H_i \cap H_j$ be the great circle in S^3 formed by the intersection of two great 2-spheres of the arrangement $\mathcal{A}(U)$, $1 \leqslant i < j \leqslant n$. Consider the subgraph of $\mathcal{A}(U)$ consisting of the vertices and edges of $\mathcal{A}(U)$ contained in C and at sublevel $(\leqslant k)$. This subgraph cannot cover the entire circle C, since C contains at least one pair of antipodal vertices at levels $\lfloor \frac{n-2}{2} \rfloor, \lceil \frac{n-2}{2} \rceil$. Thus, the connected components of this subgraph form closed intervals, which we call k-arcs of $\mathcal{A}(U)$. We denote by $\Lambda_k(U)$ the number of k-arcs in $\mathcal{A}(U)$, and we define $\lambda_k(U, u_0)$ as the number of k-arcs of $\mathcal{A}(U)$ that are completely contained in the open hemisphere H_0^-.

Our main technical result (which by the preceding discussion implies both Lemma 12 and Theorem 13) is the following.

Proposition 28. Consider a vector configuration $U = \{u_1, \ldots, u_n\} \subset \mathbb{R}^4$ and a great semicircle γ in S^3 with endpoints $u_{\pm\infty}$ such that the properties of Lemma 26 are satisfied. Let $0 \leqslant k \leqslant \lfloor \frac{n-4}{2} \rfloor$„ and let $u_0 \in \gamma$ such that $U \sqcup \{u_0\}$ is in general

position. Then

$$g_{1,k}(U/u_0) = \lambda_k(U,u_0) \qquad \text{and} \qquad g^*_{1,k}(U/u_0) = (k+1)n - 3\binom{k+2}{2} - \lambda_k(U,u_0)$$

To prove Proposition 28, we consider how the vector configuration U/u_0 changes as u_0 moves continuously along γ. In terms of the dual arrangement, a mutation of U/u_0 occurs exactly when the great 2-sphere H_0 dual to u_0 passes over a vertex v of the arrangement $\mathcal{A}(U)$, whose level we will denote by k (concurrently, H_0 passes over the antipodal vertex $-v$ of level $n-3-k$, so we will assume that $k \leqslant \lfloor \frac{n-3}{2} \rfloor$). The vertex v is incident to exactly three edges of $\mathcal{A}(U)$ at level k (each of which corresponds to a k-arc with endpoint v). Consider the moment right before or right after the mutation when v lies in the negative halfspace H_0^-, and let $j \in \{0,1,2,3\}$ denote the number of level k edges of $\mathcal{A}(U)$ incident to v that are contained in the negative hemisphere H_0^-. We call this a transition of Type $(j,k)^+$ or $(j,k)^-$ depending on whether v lies in H_0^+ or in H_0^- after the mutation.

Lemma 29. Let $0 \leqslant k \leqslant \lfloor \frac{n-4}{2} \rfloor$, and suppose that u_0 moves continuously along γ.

1. There are no transitions of Type $(2,k)$ or $(3,k)$.
2. if $(U,u_0) \to (U,u_0')$ is a transition of Type $(0,k)^{\pm}$ then λ_k and $g_{1,k}$ remain unchanged (as well as all other $g_{j,\ell}$, $\ell \neq k$), and $g_{0,k}(U/u_0 \to U,u_0') = \mp 1$;
3. if $(U,u_0) \to (U,u_0')$ is a transition of Type $(1,k)^{\pm}$ then $g_{0,k}$ (as well as all other $g_{j,\ell}$, $\ell \neq k$) remain unchanged, $\lambda_k(U,u_0') = \lambda_k(U,u_0) \mp 1$, and $g_{0,k}(U/u_0 \to U,u_0') = \mp 1$.

Corollary 30. For $0 \leqslant k \leqslant \lfloor \frac{n-4}{2} \rfloor$ and pair $u_0, u_0' \in \gamma$,

$$\lambda_k(U,u_0') - \lambda_k(U,u_0) = g_{1,k}(U/u_0') - g_{1,k}(U/u_0) = g_{1,k}((U/u_0) \to (U/u_0'))$$

Proof of Lemma 29. Suppose there is a transition of Type $(2,k)^{\pm}$ or Type $(3,k)^{\pm}$. Let v be the corresponding vertex at level k in $\mathcal{A}(U)$ and consider the moment just before or after the transition when v is in the negative hemisphere H_0^-. Without loss of generality, let H_1, H_2, H_3 be the three great 2-spheres of $\mathcal{A}(U)$ intersecting in v and in the antipodal vertex $-v$, which is at level $n-3-k$. Since

$k \leqslant \lfloor \frac{n-4}{2} \rfloor$, we have $k < n-3-k$, hence there is another hemisphere of $\mathcal{A}$, say H_4, such that $v \in H_4^+$ and $-v \in H_4^-$. If the transition is of Type $(3,k)^{\pm}$, this implies that the intersection of the hemispheres $\bigcap_{i=0}^4 H_0^+$ is empty, which is impossible since $\{u_0\} \sqcup U$ is a pointed configuration. Suppose then that the transition is of Type $(2,k)^{\pm}$. Consider the unique level k edge of $\mathcal{A}(U)$ incident to v that is not contained in H_0^-. This edge lies in the intersection of two out of the three great 2-spheres of $\mathcal{A}(U)$ intersecting in v, say in the intersection $H_2 \cap H_3$. It follows that the intersection $H_0^+ \cap H_1^- \cap \bigcap_{i=2}^4 H_i^+$ is empty. This means that there is a linear dependency between the vectors u_0, u_1, u_2, u_3, u_4 with exactly one negative coefficient, corresonding to u_1. This contradicts Property 3 of Lemma 26 (one of the defining properties of a special pair). Only the three k-arcs ending at v can contribute to a change in λ_k during a transition. If the transition is of Type $(0,k)^{\pm}$, then the three k-arcs incident to v intersect H_0^+ both before and after the transition, so λ_k does not change. If the transition is of Type $(1,k)^{\pm}$, then there is a unique k-arc incident to v that intersects H_0^-, and this k-arc contributes 1 to λ_k when $v \in H_0^-$ and 0 otherwise. This shows that λ_k changes as stated. For the changes in $g_{j,k}(U/u_0)$, $j = 0, 1$, see Figures 4 and 5. □

We will need the following well-known fact (see, e.g., [40, Corollary 8]).

Lemma 31. Let $W \subset \mathbb{R}^r$ be a neighborly arrangment of n vectors, $r \in \{3, 4\}$, and let $\mathcal{A}(W)$ be the polar dual arrangement in S^d, $d = r-1$. Then for $0 \leqslant k \leqslant n-r+1$, the number $f_{d,k}(W)$ of vertices of $\mathcal{A}(W)$ at level k equals $f_{2,k}(W) = n$ for $r = 3$, and $f_{3,k}(W) = 2n(k+1) - 4\binom{k+2}{2} = 2(k+1)(n-k-2)$ for $r = 4$.

Proof of Proposition 28. For every k-arc α of $\mathcal{A}$, the two endpoints of α are vertices of $\mathcal{A}$ at level k (while the interior of a k-arc consists of edges at sublevel $(\leqslant k)$. Conversely, every vertex at level k is an endpoint of exactly three k-arcs (corresponding to the three edges at level k incident to that vertex). Thus, By Lemma 31,

$$\Lambda_k(U) = 3(k+1)(n-k-2) = 3(k+1)n - 6\binom{k+2}{2} \tag{14}$$

Consider the great 2-sphere $H_\infty := \{x \in S^3 \colon \langle u_{+\infty}, x\rangle = 0\}$ and the two open hemispheres $H_\infty^+ := \{x \in S^3 \colon \langle u_{+\infty}, x\rangle > 0\}$ and $H_\infty^- :=$

$\{x \in S^3 \colon \langle u_{+\infty}, x\rangle < 0\}$. The 2-dimensional arrangement $\mathcal{A} \cap H_\infty$ is polar dual to the neighborly configuration $U/u_{+\infty} = U/u_{+\infty}$, hence the number of vertices of $\mathcal{A} \cap H_\infty$ at sublevel $(\leqslant k)$ equals $(k+1)n$. Consider a k-arc α of $\mathcal{A}$. Then either $\alpha \subset H_\infty^+$, or $\alpha \subset H_\infty^-$, or α intersects H_∞; k-arcs of the first two kinds are counted by $\lambda_k(U, u_{-\infty})$ and $\lambda_k(U, u_{+\infty})$, respectively, while k-arcs of the third kind correspond bijectively to vertices at sublevel $(\leqslant k)$ in $\mathcal{A} \cap H_\infty$. Therefore,

$$\Lambda_k(U) = 3(k+1)n - 6\binom{k+2}{2} = \lambda_k(U, u_{-\infty}) + \lambda_k(U, u_{+\infty}) + (k+1)n$$

hence

$$\lambda_k(U, u_{-\infty}) + \lambda_k(U, u_{+\infty}) = 2(k+1)n + 6\binom{k+2}{2} \tag{15}$$

Consider now a unit vector $u_0 \in \gamma$. By applying Corollary 30 with $u_0' = u_{+\infty}$ and using that $U/u_{+\infty}$ is neighborly, we get

$$g_{1,k}^*(U/u_0) = g_{1,k}((U/u_0) \to (U/u_{+\infty})) = \lambda_k(U, u_{+\infty}) - \lambda_k(U, u_0) \tag{16}$$

In particular, since $U/u_{+\infty}$ and $U/u_{-\infty}$ are identical, Equations (15) and (16) imply

$$\lambda_k(U, u_{+\infty}) = \lambda_k(U, u_{-\infty}) = (k+1)n + 3\binom{k+2}{2} \tag{17}$$

Substituting this back into (16), we get, for every $u_0 \in \gamma$,

$$g_{1,k}^*(U/u_0) = (k+1)n - 3\binom{k+2}{2} - \lambda_k(U, u_0) \tag{18}$$

which proves the second half of Proposition 28. It remains to show that $g_{1,k}^*(V) = (k+1)n - 3\binom{k+2}{2}$ for $0 \leqslant k \leqslant \lfloor \frac{n-4}{2} \rfloor$ whenever V is coneighborly. This follows from the following observation: If $V = S - o = U/u_0$ is coneighborly, then every vertex v of $\mathcal{A}(U)$ at level k must be contained in the positive hemisphere H_0^+ (otherwise, v would correspond to an affine hyperplane H in $\mathbb{R}^3$ that is spanned by three points of S and such that $o \in H^-$ and $|H^- \cap S| \leqslant k$; but then we could translate H towards o and obtain a hyperplane through o that still contains at most $\lfloor \frac{n-4}{2} \rfloor = \lfloor \frac{n-2}{2} \rfloor - 1$ points of S, contradicting coneighborliness of $V = S - o$). It follows from the observation that all k-arcs of $\mathcal{A}(U)$ are contained in H_0^+, hence $\lambda_k(U, u_0) = 0$, which together with (18) implies $g_{1,k}^*(V) = (k+1)n - 3\binom{k+2}{2}$ as we wanted to show. □

We remark that the notion of k-arcs in arrangements in S^3 (and in particular Equation (14)) have been used before in related contexts, see, e.g., [19, 32].

A The Remaining Calculation for the Spherical Arc Crossing Number

We want to show that

$$\sum_{k=0}^{\lfloor\frac{n-4}{2}\rfloor}(n-3-2k)\left((k+1)n-3\binom{k+2}{2}\right)=\binom{n}{4}-X(n) \tag{10}$$

where

$$X(n)=\frac{1}{4}\left\lfloor\frac{n}{2}\right\rfloor\left\lfloor\frac{n-1}{2}\right\rfloor\left\lfloor\frac{n-2}{2}\right\rfloor\left\lfloor\frac{n-3}{2}\right\rfloor$$

The left-hand side of (10) equals

$$(n-3)n\sum_{k=0}^{\lfloor\frac{n-4}{2}\rfloor}(k+1)-3(n-3)\sum_{k=0}^{\lfloor\frac{n-4}{2}\rfloor}\binom{k+2}{2}-2n\sum_{k=0}^{\lfloor\frac{n-4}{2}\rfloor}k(k+1)+6\sum_{k=0}^{\lfloor\frac{n-4}{2}\rfloor}k\binom{k+2}{2} \tag{19}$$

Using the basic identity

$$\sum_{k=0}^{m}\binom{k+i}{i}=\binom{m+i+1}{i+1}$$

we see that (19) equals

$$\begin{aligned}
& n(n-3)\binom{\lfloor\frac{n}{2}\rfloor}{2}-3(n-3)\binom{\lfloor\frac{n+2}{2}\rfloor}{3}-4n\binom{\lfloor\frac{n}{2}\rfloor}{3}+18\binom{\lfloor\frac{n+2}{2}\rfloor}{4}\\
= \quad & \left\lfloor\frac{n}{2}\right\rfloor\left\lfloor\frac{n-2}{2}\right\rfloor\underbrace{\left(\frac{1}{2}n(n-3)-\frac{1}{2}(n-3)\left\lfloor\frac{n+2}{2}\right\rfloor-\frac{2}{3}n\left\lfloor\frac{n-4}{2}\right\rfloor+\frac{3}{4}\left\lfloor\frac{n+2}{2}\right\rfloor\left\lfloor\frac{n-4}{2}\right\rfloor\right)}_{(*)}\\
= \quad & \left\lfloor\frac{n}{2}\right\rfloor\left\lfloor\frac{n-2}{2}\right\rfloor\underbrace{\left(\frac{2}{3}\left(\left\lfloor\frac{n-1}{2}\right\rfloor+\frac{1}{2}\right)\left(\left\lfloor\frac{n-3}{2}\right\rfloor+\frac{1}{2}\right)-\frac{1}{4}\left\lfloor\frac{n-1}{2}\right\rfloor\left\lfloor\frac{n-3}{2}\right\rfloor\right)}_{(**)}
\end{aligned}$$

where the last step amounts to the assertion that the expressions (*) and (**) are equal as quadratic polynomials in n, which is easy

to check. Equation (10) then follows from the identify

$$\binom{n}{4} = \frac{2}{3}\left\lfloor\frac{n}{2}\right\rfloor\left\lfloor\frac{n-2}{2}\right\rfloor\left(\left\lfloor\frac{n-1}{2}\right\rfloor+\frac{1}{2}\right)\left(\left\lfloor\frac{n-3}{2}\right\rfloor+\frac{1}{2}\right)$$

which is likewise easy to verify. □

Author context for the two equation references

Throughout this paper, let $r = d+1 \geqslant 1$, and let S^d be the unit sphere in $\mathbb{R}^r$. We denote the standard inner product in $\mathbb{R}^r$ by $\langle\cdot,\cdot\rangle$, and write $\operatorname{sgn}(x) \in \{-1, 0, +1\}$ for the sign of a real number $x \in \mathbb{R}$.

Observation 32. By antipodal symmetry $F \leftrightarrow -F$ of $\mathcal{F}(V)$,

$$f_{s,t}(V) = f_{s,n-s-t}(V) \quad \text{for all } s \text{ and } t\text{; equivalently,} \quad f_V(x,y) = y^n f_V(\tfrac{x}{y}, \tfrac{1}{y}) \tag{20}$$

Theorem 33 (Dehn–Sommerville Relations for Levels in Simple Arrangements)**.** Let $V \in \mathbb{R}^{r\times n}$ be a vector configuration in general position. Then

$$f_V(x,y) = (-1)^d f_V(-(x+y+1), y) \tag{21}$$

Equivalently (by comparing coefficients), for $0 \leqslant s \leqslant d$ and $0 \leqslant t \leqslant n$,

$$f_{s,t}(V) = \sum_j \sum_\ell (-1)^{d-j}\binom{j}{s}\binom{j-s}{t-\ell} f_{j,\ell}(V) \tag{22}$$

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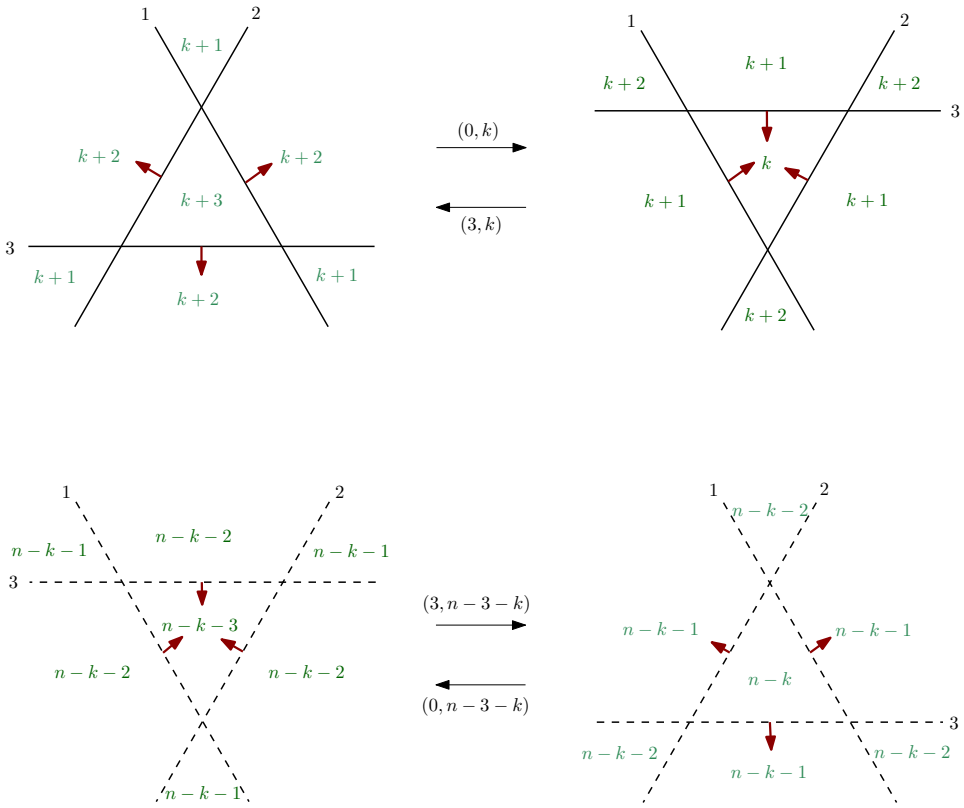


Figure 1: A mutation between an antipodal pair of triangles $\sigma, -\sigma$ of types $(3,k)$ and $(0,n-3-k)$ (left column top and bottom) and an antipodal pair $\tau, -\tau$ of triangles of types $(0,k)$ and $(3,n-3-k)$ (right column top and bottom). The horizontal arrows are marked with the types of the appearing triangles as we move between the left and the right column. The labels in the full-dimensional cells indicate their levels, and the little arrows indicate positive hemispheres.

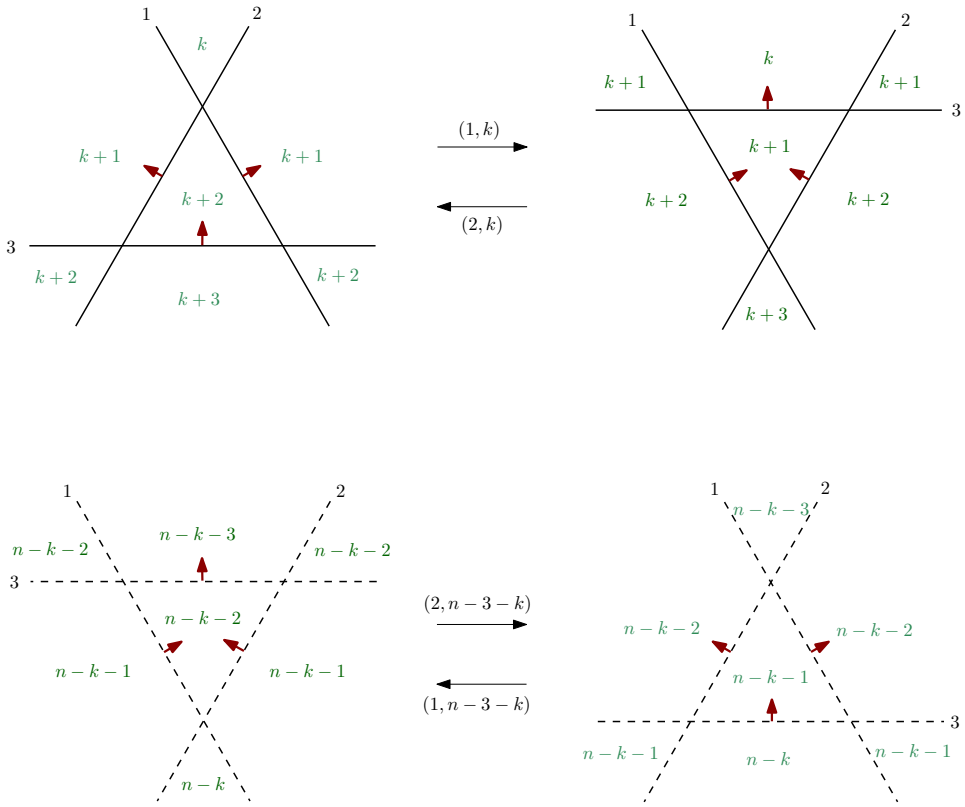


Figure 2: A mutation between an antipodal pair of triangles $\sigma, -\sigma$ of types $(2,k)$ and $(1,n-3-k)$ (left column) and an antipodal pair $\tau, -\tau$ of triangles of types $(1,k)$ and $(2,n-3-k)$ (right column).

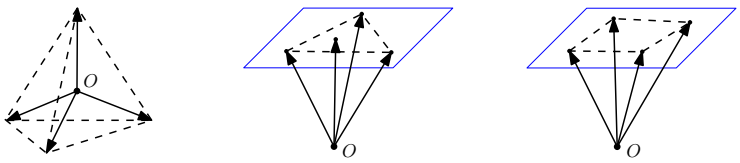


Figure 3: The three combinatorial types of four vectors in general position in $\mathbb{R}^3$.

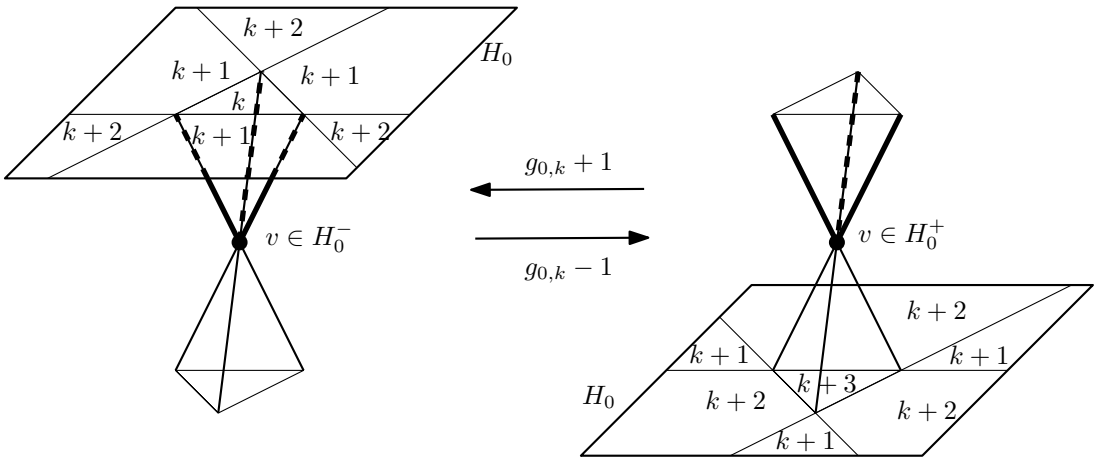


Figure 4: Transitions of Type $(0,k)^+$ (left to right) and $(0,k)^-$ (right to left) and the change in $g_{0,k}(U/u_0)$; the labels $k, k+1, k+2, k+3$ show the levels of the 2-cells in $\mathcal{A} \cap H_0$.

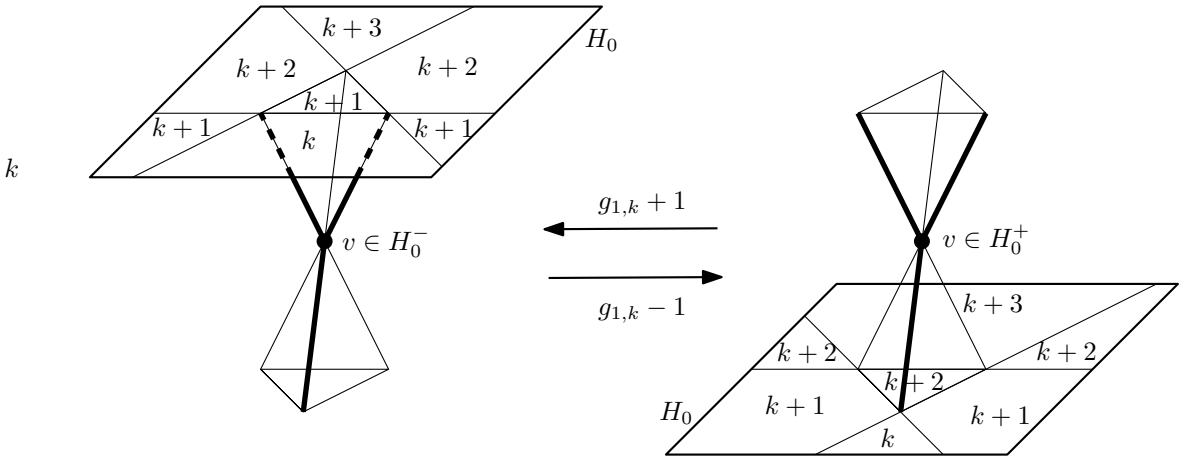


Figure 5: Transitions of Type $(1,k)^+$ (left to right) and $(1,k)^-$ (right to left) and the corresponding change in $g_{1,k}(U/u_0)$; the labels $k, k+1, k+2, k+3$ show the levels of the 2-cells in $\mathcal{A} \cap H_0$.